

***Test for improvements on rate of combustion of  
biofuel (vegetable oil) using silver nanoparticles  
(AgNPs) and carbon nanotubes (CNTs)***

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## Abstract

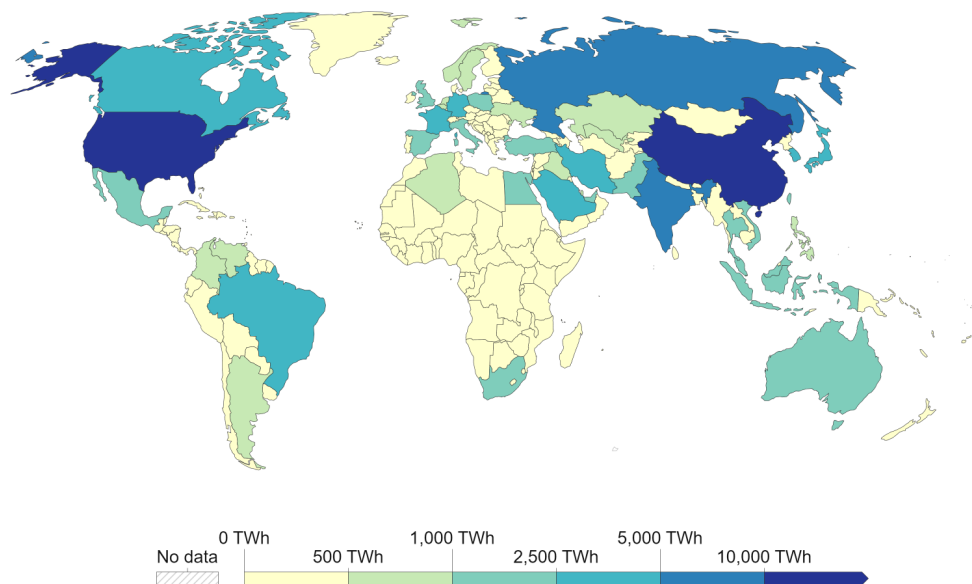
To cope with the possible scarcity of fossil fuels within a few decades, biofuel was introduced as a promising alternative, enabling a transition for industries that use fossil fuel and vehicles to adapt without a lot of modifications, or even a complete update in components, or machines used. However, in terms of energy density, biofuels generally have lower energy content compared to fossil fuels, which can result in reduced mileage and energy efficiency when used in conventional internal combustion engines. Therefore, silver nanoparticles (AgNPs) and carbon nanotubes (CNTs) have been attempted to use as a catalyst of the combustion of biofuels, enabling similar efficiency as fossil fuel combustions. Dynamic Light Scattering (DLS) is used to characterise AgNPs and CNTs in their suspension, determining the size distribution profile of them. Time of combustion of samples added certain amount of nanoparticles of biofuel (vegetable oil) is recorded with regard of a comparison sample without nanoparticles to determine performance of the catalysts.

## Society Problem

### Primary energy consumption, 2021

Primary energy<sup>1</sup> consumption is measured in terawatt-hours (TWh).

Our World  
in Data



Source: BP Statistical Review of World Energy; and EIA

OurWorldInData.org/energy • CC BY

Note: Data includes only commercially-traded fuels (coal, oil, gas), nuclear and modern renewables. It does not include traditional biomass.

**1. Primary energy:** Primary energy is the energy as it is available as resources – such as the fuels that are burnt in power plants – before it has been transformed. This relates to the coal before it has been burned, the uranium, or the barrels of oil. Primary energy includes energy that is needed by the end user, plus inefficiencies and energy that is lost when raw resources are transformed into a useable form. You can read more on the different ways of measuring energy in our article.

Fig. 1

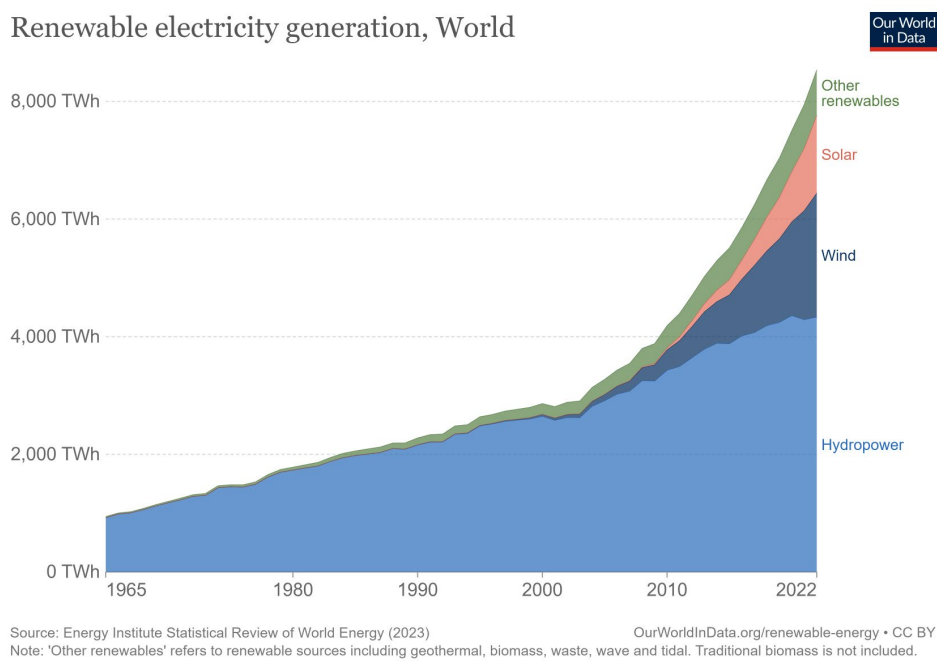


Fig. 2

As shown in Fig. 1, the world is currently consuming energy in a very uneven way, among those only less than  $\frac{1}{2}$  is provided with renewable methods.

### Years of fossil fuel reserves left, 2020

Years of global coal, oil and natural gas left, reported as the reserves-to-product (R/P) ratio which measures the number of years of production left based on known reserves and present annual production levels. Note that these values can change with time based on the discovery of new reserves, and changes in annual production.

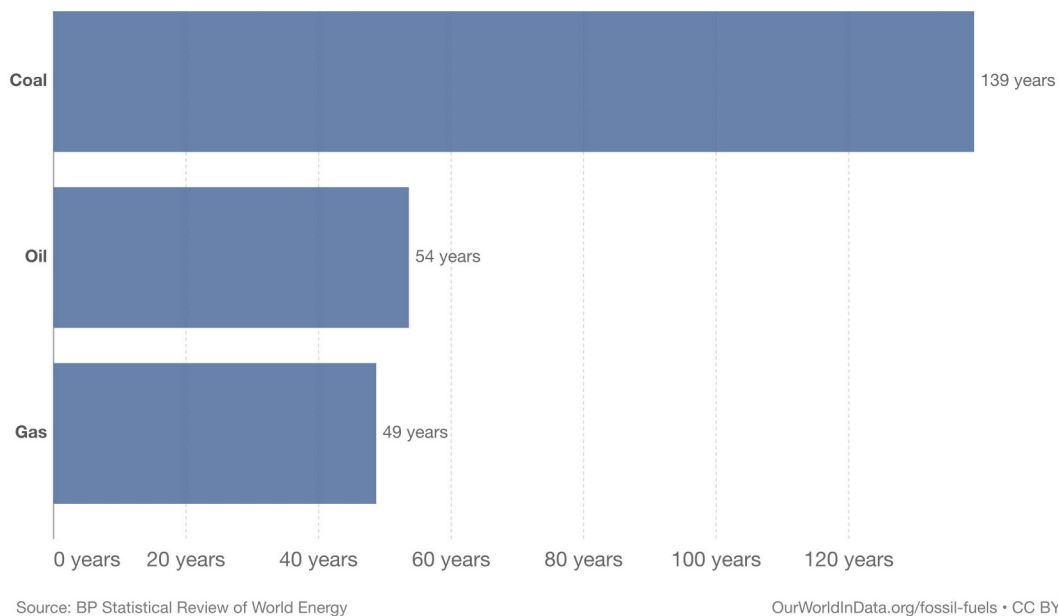


Fig. 3

*'Formed millions of years ago, yet only used for around 200 years, fossil fuel reserves are emptying very quickly. It is also obvious that the exact date of running out of these fuels remains unknown. It is because we continue to discover new reserves, however, the number of new reserves is low: they cannot meet our population's needs with the current and expected future levels of usage.'* (MET Group)

As the fossil fuel reserve is running low, alternatives have been proposed to use including gaseous fuels such as hydrogen, natural gas, and propane; alcohols such as ethanol, methanol, and butanol; vegetable and waste-derived oils; and electricity. These fuels may be used in a dedicated system that burns a single fuel, or in a mixed system with other fuels including traditional gasoline or diesel, such as in hybrid-electric or flexible fuel vehicles.

Biofuel can be used in both systems, giving it advantage compared with other renewable energy such as electricity. Furthermore, considering its economic and political benefits, for example, in the United States:

*'The United States became a net exporter of petroleum in 2020 with exports surpassing imports, although imports of 8.47 million barrels per day in 2021 remained an important part of balancing supply and demand for domestic and international markets. Overall, the transportation sector accounts for approximately 30% of total U.S. energy needs and 70% of U.S. petroleum consumption. Using biodiesel and other alternative fuels and advanced technologies to reduce petroleum consumption continues to strengthen national security and reduce transportation energy costs for businesses and consumers.'*

*Biodiesel is produced in the United States and used in conventional diesel engines, directly substituting for or extending supplies of traditional petroleum diesel. Soybean biodiesel has a positive energy balance, meaning that soybean biodiesel yields 4.56 units of energy for every unit of fossil energy consumed over its life cycle.'* (US EPA)

Moreover, in the application of it as vehicle fuel, biodiesel utilization leads to decreased life cycle emissions as the carbon dioxide generated during its combustion is balanced by the carbon dioxide absorbed during the growth of soybeans or other feedstocks employed in its production.

*'Life cycle analysis completed by Argonne National Laboratory found that B100 use reduces carbon dioxide emissions by 74% compared with petroleum diesel.'* (US EPA)

Biodiesel also enhances fuel lubricity and elevates the cetane number, both critical factors for the smooth functioning of diesel engines, preventing premature wear of moving parts. Unintentionally, the federal regulations aimed at reducing allowable fuel sulfur to 15 ppm and lowering aromatics content have resulted in diminished lubricity of petroleum diesel.

Biodiesel possesses the ability to enhance diesel fuel lubricity, even in small quantities. The precise amount of biodiesel needed relies on the characteristics of the diesel fuel; however, a mere 2% of biodiesel is typically ample to ensure satisfactory lubrication.

Despite those promising advantages of biofuel, they often have lower energy densities than fossil fuels, which means they contain less energy per unit volume. As a result, they may produce less power when burned in engines, leading to reduced combustion efficiency.

To counter this disadvantage, it is proposed to produce and analyse nanoparticles such as AgNPs and CNTs, to reduce the time of combustion of a certain amount of biofuel, in this specific experiment, vegetable oil.

***Experiment Procedure (i) – Production of nanoparticles and analytical data of size by Dynamic Light Scattering (DLS)***

## ***I. Silver Nanoparticles (AgNPs)***

To produce the AgNPs, 2 methods have been used, involving extraction from *CVS silver wound gel* and precipitation between silver nitrate and sodium hydroxide.

### ***i. Using CVS silver wound gel***



*Fig. 4 Demonstration photo of the silver wound gel used in experiment from CVS website*

Chamomile tea was boiled with the silver wound gel to extract silver nanoparticles from it, where chamomile terpenoids was used as a combined reducing and capping agent. (Parlinska-Wojtan et al.)



*Fig. 5 Chamomile tea boiled with silver wound gel*

Next, the mixture was stirred under the sun for 5 minutes, and stayed under the sun for 40 minutes.

The sample is then analysed using DLS.

Results shown by the following.

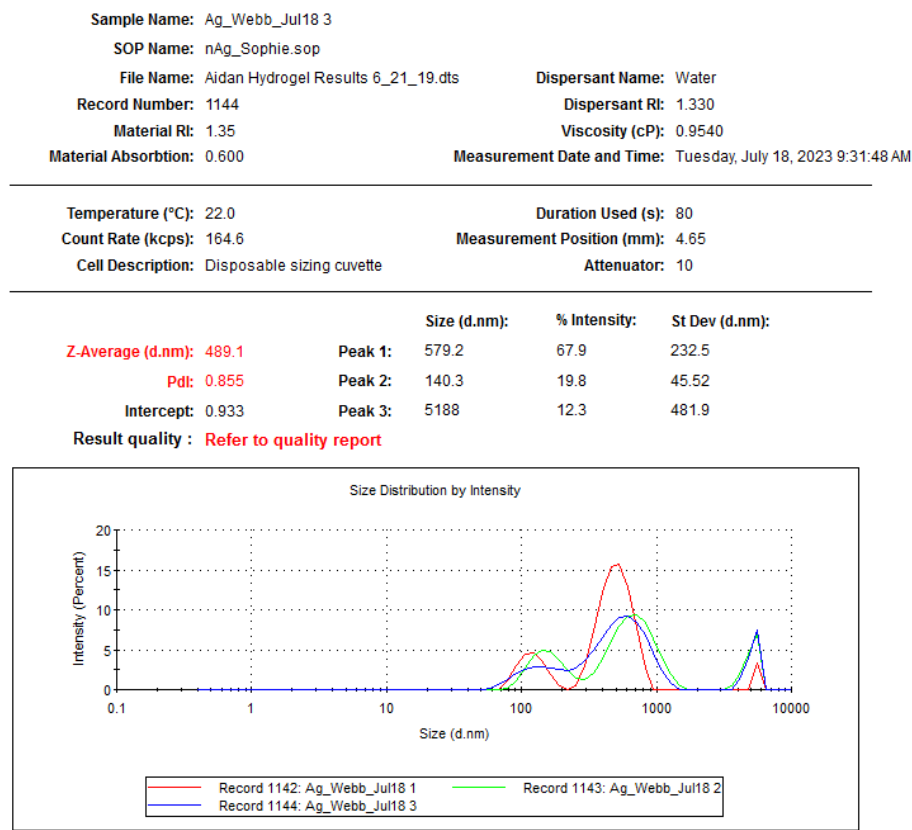
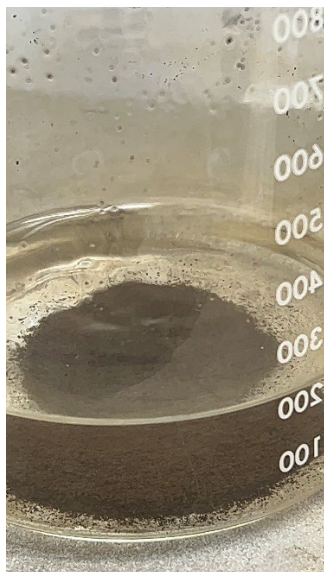


Fig. 6 DLS results from silver nanoparticles extracted from silver wound gel

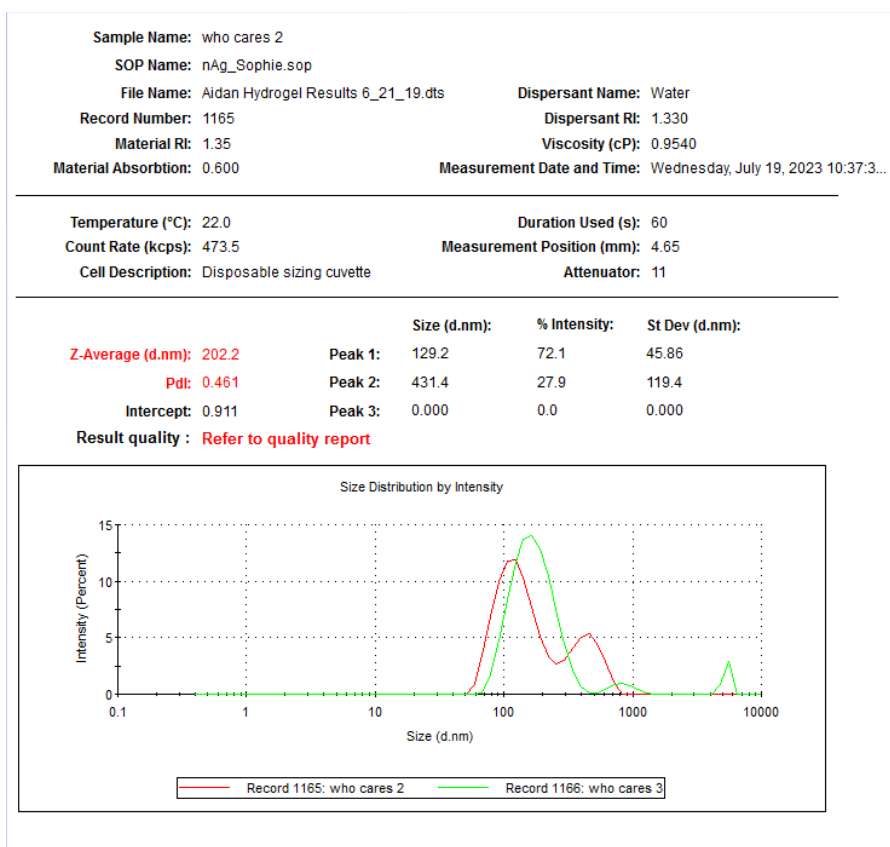
ii. ***Precipitation from reaction between silver nitrate and sodium hydroxide***

0.1  $\text{mol dm}^{-3}$  silver nitrate aqueous solution and 1  $\text{mol dm}^{-3}$  were prepared and mixed together to precipitate and produce the AgNPs needed.



*Fig. 6 Photo of AgNPs made with method ii*

The sample is then analysed using DLS, result is shown by the following.



*Fig. 7 DLS results from silver nanoparticles extracted from silver wound gel*

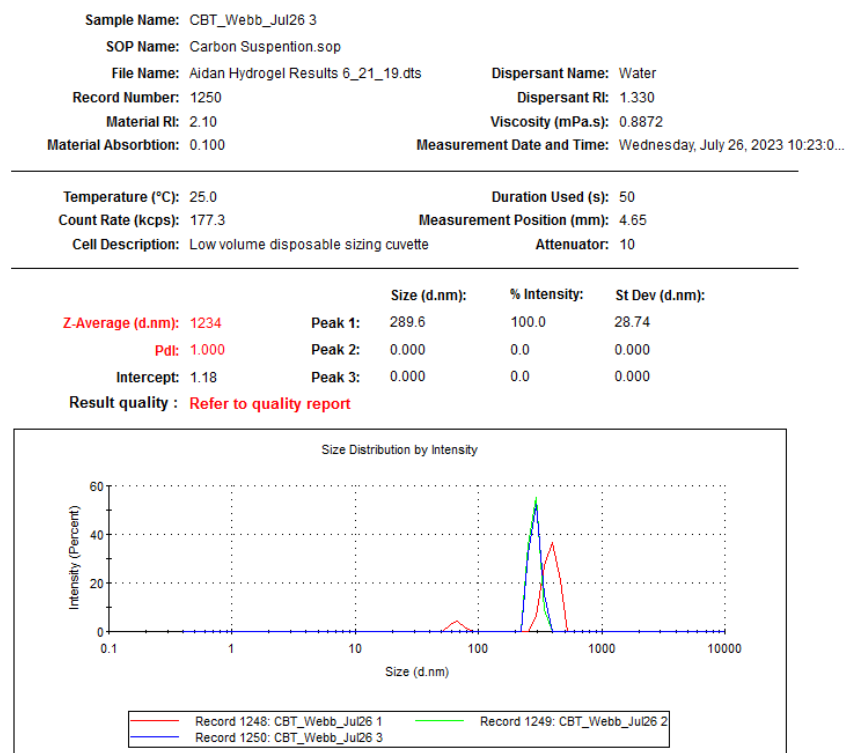
Eventually, the solution was centrifuged to extract silver nanoparticles.

Comparing the two DLS results, it is considered that method *ii* produces nanomaterial that has higher quality, due to its smaller diameter, therefore larger area of contact when used as catalyst, making it better in catalytical influence, therefore, the silver nanoparticles produced by that method was chosen to be the silver catalyst used in the combustion experiment.

## II. Carbon Nanotubes (CNTs)

The carbon nanotubes was from the teaching laboratory in Engineering Department of Brown University.

DLS results is shown in the following.



*Fig. 8 DLS results from CNTs*

As shown in chart the overall average of the CNTs is over 1000 nanometres, however, the amount of the large particles is negligible, and it was because after being stored in the lab cupboard for too long the CNTs were combining into larger particles, therefore the peak 1

with size 289.6 nanometre was more convincing and shows the CNTs have a similar size as the AgNPs.

### ***Combustion Experiment and Results***



*Fig. 8 Apparatus Photo of Combustion Experiment*

Three 15 cc samples – control (without any nano-catalysts), with 0.1 g AgNPs and with 0.1 g CNTs.

The combustion time was recorded, which was 3' 17", 2' 44" and 2' 28", respectively, as a mean of 3 repeated experiment for each sample.

Therefore, a 16.8 % and 24.9 % rise in combustion efficiency is recorded for AgNPs and CNTs, respectively.

## ***Conclusion***

In conclusion, the findings of this study show that AgNPs and CNTs as biofuel additives significantly improve combustion efficiency. The 16.8% and 24.9% increases in AgNPs and CNTs, respectively, highlight the promising potential of these nanomaterials in boosting the overall performance of biofuels. These discoveries bring up new avenues for optimising biofuel combustion processes, resulting in lower emissions and improved energy utilisation. As the world pursues ecologically friendly and sustainable energy options, using AgNPs and CNTs as biofuel enhancers offers enormous potential for a greener and more efficient future. More research and development in this field is required to fully realise the benefits of these nanomaterials and bring them closer to practical applications in real-world circumstances. Further researches could focus on real world applications, such as performance of the nanomaterials in a conventional petrol or diesel engine.

## ***Reference***

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